OOPJ LAB ASSIGNMENT – 11

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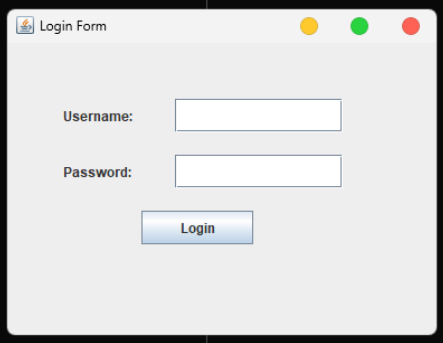
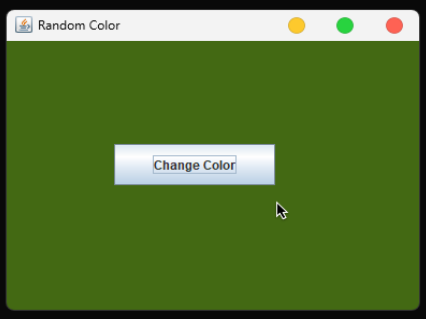
1. Write a Java Swing program that creates a login form with username and password fields and validates the input when the login button is clicked.

```java
import javax.swing.*;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;

class as_11_q_1 {
    public static void main(String[] args) {
        JFrame f = new JFrame( title: "Login Form");
        JLabel l1 = new JLabel( text: "Username:");
        JLabel l2 = new JLabel( text: "Password:");
        JTextField t1 = new JTextField();
        JPasswordField t2 = new JPasswordField();
        JButton b = new JButton( text: "Login");
        l1.setBounds( x: 50, y: 50, width: 100, height: 30);
        t1.setBounds( x: 150, y: 50, width: 150, height: 30);
        l2.setBounds( x: 50, y: 100, width: 100, height: 30);
        t2.setBounds( x: 150, y: 100, width: 150, height: 30);
        b.setBounds( x: 120, y: 150, width: 100, height: 30);
        b.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {
                String user = t1.getText();
                String pass = new String(t2.getPassword());
                if (user.equals("admin") && pass.equals("1234"))
                    JOptionPane.showMessageDialog(f, message: "Login Successful");
                else
                    JOptionPane.showMessageDialog(f, message: "Invalid Login");
            }
        });
        f.add(l1);
        f.add(t1);
        f.add(l2);
        f.add(t2);
        f.add(b);
        f.setSize( width: 400, height: 300);
        f.setLayout(null);
        f.setVisible(true);
    }
}
```



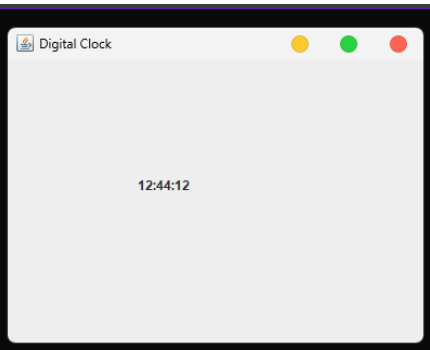
2. Design a Swing application that changes the background color of the window randomly whenever a button is pressed.

```java
import javax.swing.*;
import java.awt.*;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;

class as_11_q_2 {
    public static void main(String[] args) {
        JFrame f = new JFrame( title: "Random Color");

        JButton b = new JButton( text: "Change Color");

        b.setBounds( x: 100, y: 100, width: 150, height: 40);

        b.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {
                Color c = new Color(
                        (int) (Math.random() * 255),
                        (int) (Math.random() * 255),
                        (int) (Math.random() * 255)
                );

                f.getContentPane().setBackground(c);
            }
        });

        f.add(b);

        f.setSize( width: 400, height: 300);
        f.setLayout(null);
        f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        f.setVisible(true);
    }
}
```



3. Create a Swing program that displays the current system time and updates it every second.

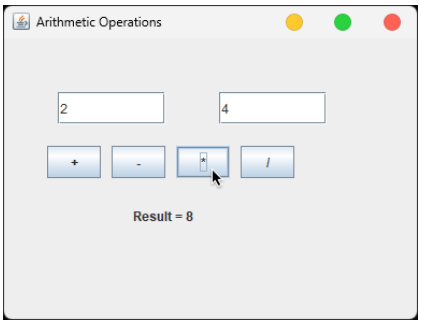
```java
import javax.swing.*;
import java.awt.event.*;
import java.text.SimpleDateFormat;
import java.util.Date;

class as_11_q_3 {
    public static void main(String[] args) {
        JFrame f = new JFrame("Digital Clock");

        JLabel l = new JLabel();

        l.setBounds(120, 100, 200, 30);

        Timer t = new Timer(1000, new ActionListener() {
            public void actionPerformed(ActionEvent e) {
                SimpleDateFormat sdf = new SimpleDateFormat("HH:mm:ss");
                l.setText(sdf.format(new Date()));
            }
        });

        t.start();

        f.add(l);

        f.setSize(400, 300);
        f.setLayout(null);
        f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        f.setVisible(true);
    }
}
```



4. Write a Swing application that accepts two numbers from text fields and performs arithmetic operations using buttons.

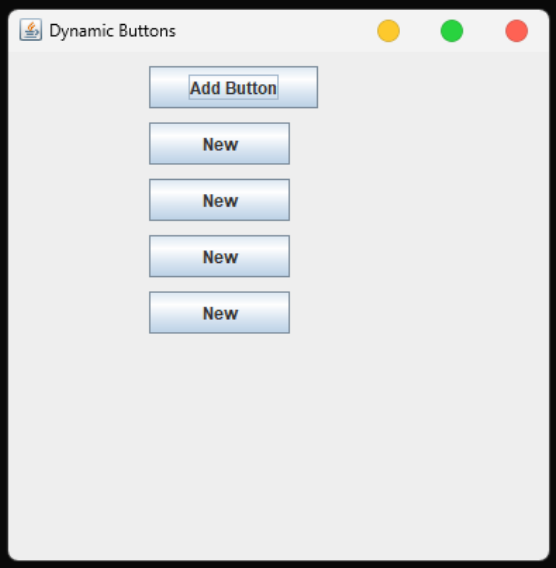
```java
import javax.swing.*;
import java.awt.event.*;
class as_11_q_4 {
    public static void main(String[] args) {
        JFrame f = new JFrame("Arithmetic Operations");
        JTextField t1 = new JTextField();
        JTextField t2 = new JTextField();
        JButton b1 = new JButton("+");
        JButton b2 = new JButton("-");
        JButton b3 = new JButton("*");
        JButton b4 = new JButton("/");
        JLabel l = new JLabel();
        t1.setBounds(50, 50, 100, 30);
        t2.setBounds(200, 50, 100, 30);
        b1.setBounds(40, 100, 50, 30);
        b2.setBounds(100, 100, 50, 30);
        b3.setBounds(160, 100, 50, 30);
        b4.setBounds(220, 100, 50, 30);
        l.setBounds(120, 150, 150, 30);
        b1.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {
                int a = Integer.parseInt(t1.getText());
                int b = Integer.parseInt(t2.getText());
                l.setText("Result = " + (a + b));
            }
        });
        b2.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {
                int a = Integer.parseInt(t1.getText());
                int b = Integer.parseInt(t2.getText());
                l.setText("Result = " + (a - b));
            }
        });
        b3.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {
                int a = Integer.parseInt(t1.getText());
                int b = Integer.parseInt(t2.getText());
                l.setText("Result = " + (a * b));
            }
        });
```

```java
        b4.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {
                int a = Integer.parseInt(t1.getText());
                int b = Integer.parseInt(t2.getText());
                l.setText("Result = " + (a / b));
            }
        });
        f.add(t1); f.add(t2); f.add(b1); f.add(b2); f.add(b3); f.add(b4); f.add(l);
        f.setSize(400, 300);
        f.setLayout(null);
        f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        f.setVisible(true);
    }
}
```



5. Design a GUI where clicking a button dynamically adds new buttons to the window.

```java
import javax.swing.*;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;

class as_11_q_5 {
    static int y = 50;  no usages

    public static void main(String[] args) {
        JFrame f = new JFrame( title: "Dynamic Buttons");
        JButton add = new JButton( text: "Add Button");
        add.setBounds( x: 100,  y: 10,  width: 120,  height: 30);
        add.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {
                JButton b = new JButton( text: "New");
                b.setBounds( x: 100, y,  width: 100,  height: 30);
                y += 40;
                f.add(b);
                f.repaint();
            }
        });
        f.add(add);
        f.setSize( width: 400,  height: 400);
        f.setLayout(null);
        f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        f.setVisible(true);
    }
}
```

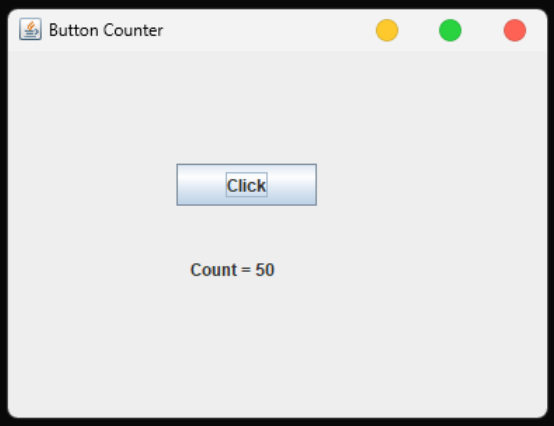


6. Write a Swing program where a label moves across the screen when a button is pressed.

```java
import javax.swing.*;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;

class as_11_q_6 {
    static int x = 50;  no usages
    public static void main(String[] args) {
        JFrame f = new JFrame( title: "Moving Label");
        JLabel l = new JLabel( text: "Moving");
        JButton b = new JButton( text: "Move");
        l.setBounds(x,  y: 100,  width: 100,  height: 30);
        b.setBounds( x: 120,  y: 180,  width: 100,  height: 30);
        b.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {
                x += 20;
                l.setBounds(x,  y: 100,  width: 100,  height: 30);
                f.repaint();
            }
        });
        f.add(l);
        f.add(b);
        f.setSize( width: 400,  height: 300);
        f.setLayout(null);
        f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        f.setVisible(true);
    }
}
```

7. Create a Swing application that counts how many times a button has been clicked.

```java
import javax.swing.*;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;

class as_11_q_7 {
    static int count = 0;  no usages

    public static void main(String[] args) {
        JFrame f = new JFrame( title: "Button Counter");

        JButton b = new JButton( text: "Click");
        JLabel l = new JLabel( text: "Count = 0");

        b.setBounds( x: 120,  y: 80,  width: 100,  height: 30);
        l.setBounds( x: 130,  y: 140,  width: 100,  height: 30);

        b.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {
                count++;
                l.setText("Count = " + count);
            }
        });

        f.add(b);
        f.add(l);

        f.setSize( width: 400,  height: 300);
        f.setLayout(null);
        f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        f.setVisible(true);
    }
}
```

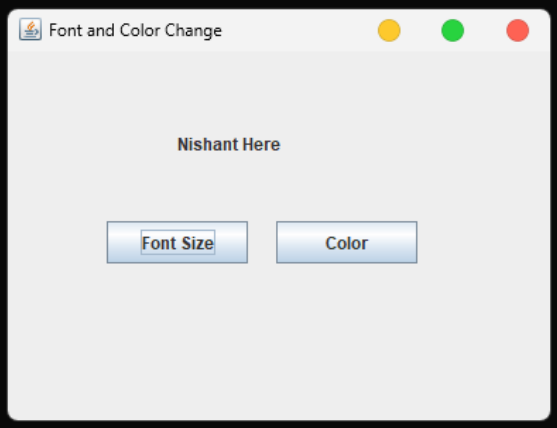


8. Write a Swing program that enables or disables buttons depending on user interaction.

```java
import javax.swing.*;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;

class as_11_q_8 {
    public static void main(String[] args) {
        JFrame f = new JFrame( title: "Enable Disable Buttons");
        JButton target = new JButton( text: "Target");
        JButton disable = new JButton( text: "Disable");
        JButton enable = new JButton( text: "Enable");
        target.setBounds( x: 130,  y: 50,  width: 100,  height: 30);
        disable.setBounds( x: 70,  y: 120,  width: 100,  height: 30);
        enable.setBounds( x: 190,  y: 120,  width: 100,  height: 30);
        disable.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {
                target.setEnabled(false);
            }
        });
        enable.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {
                target.setEnabled(true);
            }
        });
        f.add(target);
        f.add(disable);
        f.add(enable);
        f.setSize( width: 400,  height: 300);
        f.setLayout(null);
        f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        f.setVisible(true);
    }
}
```

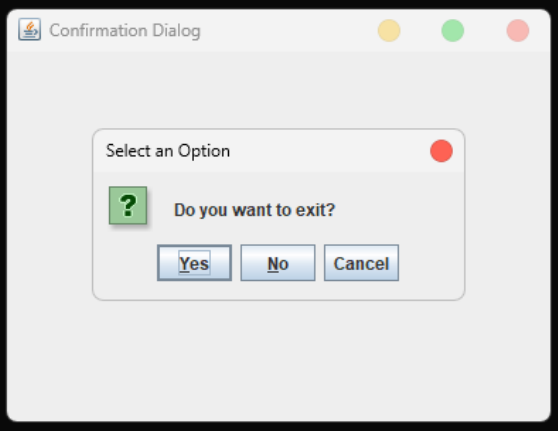
9. Design a GUI that changes the font size and color of text dynamically based on user selection.

```java
import javax.swing.*;
import java.awt.*;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;
class as_11_q_9 {
    public static void main(String[] args) {
        JFrame f = new JFrame( title: "Font and Color Change");
        JLabel l = new JLabel( text: "Nishant Here");
        JButton b1 = new JButton( text: "Font Size");
        JButton b2 = new JButton( text: "Color");
        l.setBounds( x: 120, y: 50, width: 200, height: 30);
        b1.setBounds( x: 70, y: 120, width: 100, height: 30);
        b2.setBounds( x: 190, y: 120, width: 100, height: 30);
        b1.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {
                l.setFont(new Font( name: "Arial", Font.BOLD, size: 24));
            }
        });
        b2.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {
                l.setForeground(Color.RED);
            }
        });
        f.add(l);
        f.add(b1);
        f.add(b2);
        f.setSize( width: 400, height: 300);
        f.setLayout(null);
        f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        f.setVisible(true);
    }
}
```



10. Write a Swing application that displays a confirmation dialog before closing the window.

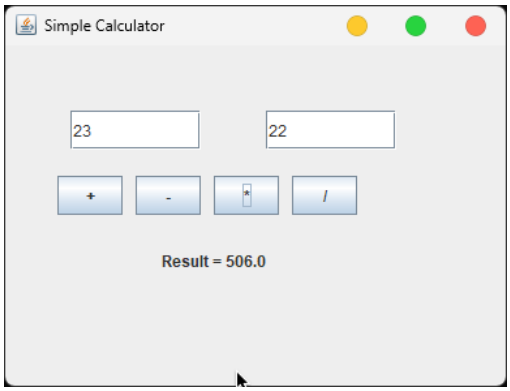
```java
import javax.swing.*;
import java.awt.event.WindowAdapter;
import java.awt.event.WindowEvent;
class as_11_q_10 {
    public static void main(String[] args) {
        JFrame f = new JFrame( title: "Confirmation Dialog");
        f.setSize( width: 400, height: 300);
        f.setLayout(null);
        f.setDefaultCloseOperation(JFrame.DO_NOTHING_ON_CLOSE);
        f.addWindowListener(new WindowAdapter() {
            public void windowClosing(WindowEvent e) {
                int x = JOptionPane.showConfirmDialog(
                        f,
                        message: "Do you want to exit?"
                );
                if (x == JOptionPane.YES_OPTION)
                    f.dispose();
            }
        });
        f.setVisible(true);
    }
}
```



11. Create a Swing program that implements a simple calculator supporting +, −, ×, ÷ operations.

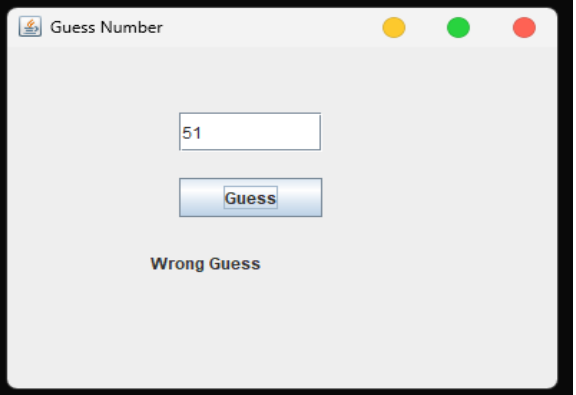
```java
import javax.swing.*;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;

class as_11_q_11 {
    public static void main(String[] args) {
        JFrame f = new JFrame("Simple Calculator");
        JTextField t1 = new JTextField();
        JTextField t2 = new JTextField();
        JButton add = new JButton("+");
        JButton sub = new JButton("-");
        JButton mul = new JButton("*");
        JButton div = new JButton("/");
        JLabel l = new JLabel();
        t1.setBounds(50, 50, 100, 30);
        t2.setBounds(200, 50, 100, 30);
        add.setBounds(40, 100, 50, 30);
        sub.setBounds(100, 100, 50, 30);
        mul.setBounds(160, 100, 50, 30);
        div.setBounds(220, 100, 50, 30);
        l.setBounds(120, 150, 150, 30);
        add.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {
                double a = Double.parseDouble(t1.getText());
                double b = Double.parseDouble(t2.getText());
                l.setText("Result = " + (a + b));
            }
        });
        sub.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {
                double a = Double.parseDouble(t1.getText());
                double b = Double.parseDouble(t2.getText());
                l.setText("Result = " + (a - b));
            }
        });
        mul.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {
                double a = Double.parseDouble(t1.getText());
                double b = Double.parseDouble(t2.getText());
```

```java
                l.setText("Result = " + (a * b));
            }
        });
        div.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {
                double a = Double.parseDouble(t1.getText());
                double b = Double.parseDouble(t2.getText());
                l.setText("Result = " + (a / b));
            }
        });
        f.add(t1);
        f.add(t2);
        f.add(add);
        f.add(sub);
        f.add(mul);
        f.add(div);
        f.add(l);
        f.setSize(400, 300);
        f.setLayout(null);
        f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        f.setVisible(true);
    }
```



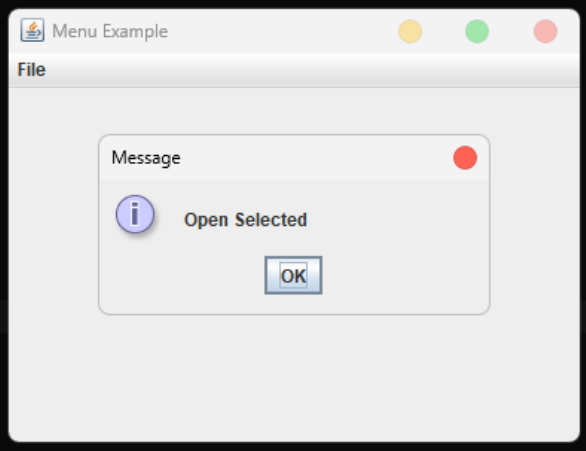
12. Design a Swing application that generates a random number and asks the user to guess it.

```java
import javax.swing.*;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;

class as_11_q_12 {
    public static void main(String[] args) {
        JFrame f = new JFrame("Guess Number");
        int num = (int) (Math.random() * 10) + 1;
        JTextField t = new JTextField();
        JButton b = new JButton("Guess");
        JLabel l = new JLabel();
        t.setBounds(120, 50, 100, 30);
        b.setBounds(120, 100, 100, 30);
        l.setBounds(100, 150, 200, 30);
        b.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {
                int g = Integer.parseInt(t.getText());
                if (g == num)
                    l.setText("Correct Guess");
                else
                    l.setText("Wrong Guess");
            }
        });
        f.add(t);
        f.add(b);
        f.add(l);
        f.setSize(400, 300);
        f.setLayout(null);
        f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        f.setVisible(true);
    }
}
```



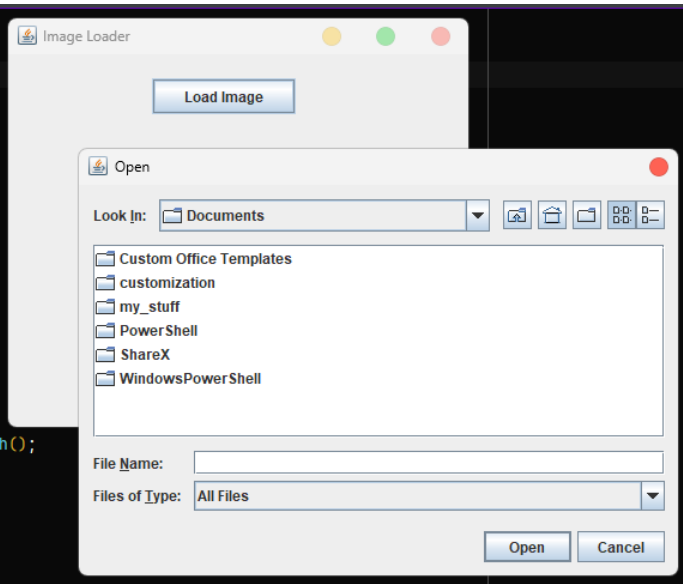
13. Write a program that creates a dynamic menu bar where menu items trigger different actions.

```java
import javax.swing.*;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;
class as_11_q_13 {
    public static void main(String[] args) {
        JFrame f = new JFrame( title: "Menu Example");
        JMenuBar mb = new JMenuBar();
        JMenu m = new JMenu( s: "File");
        JMenuItem i1 = new JMenuItem( text: "Open");
        JMenuItem i2 = new JMenuItem( text: "Exit");
        i1.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {
                JOptionPane.showMessageDialog(f, message: "Open Selected");
            }
        });
        i2.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {
                System.exit( status: 0);
            }
        });
        m.add(i1);
        m.add(i2);
        mb.add(m);
        f.setJMenuBar(mb);
        f.setSize( width: 400, height: 300);
        f.setLayout(null);
        f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        f.setVisible(true);
    }
}
```



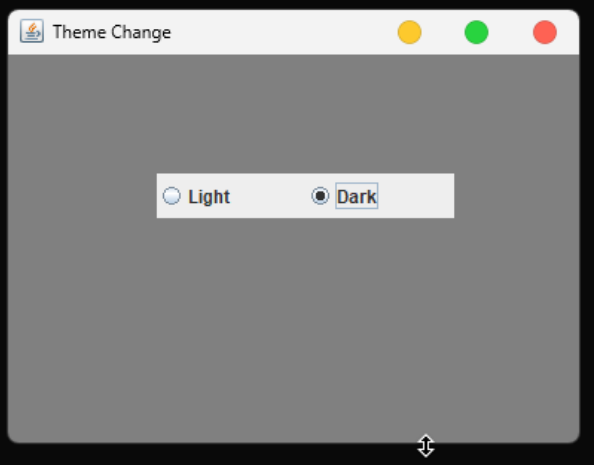
14. Design a GUI that loads and displays an image selected by the user.

```java
import javax.swing.*;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;

class as_11_q_14 {
    public static void main(String[] args) {
        JFrame f = new JFrame( title: "Image Loader");
        JButton b = new JButton( text: "Load Image");
        JLabel l = new JLabel();
        b.setBounds( x: 120, y: 20, width: 120, height: 30);
        l.setBounds( x: 50, y: 70, width: 300, height: 200);
        b.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {
                JFileChooser fc = new JFileChooser();
                int x = fc.showOpenDialog(f);
                if (x == JFileChooser.APPROVE_OPTION) {
                    String path = fc.getSelectedFile().getAbsolutePath();
                    ImageIcon img = new ImageIcon(path);
                    l.setIcon(img);
                }
            }
        });
        f.add(b);
        f.add(l);
        f.setSize( width: 400, height: 350);
        f.setLayout(null);
        f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        f.setVisible(true);
    }
}
```



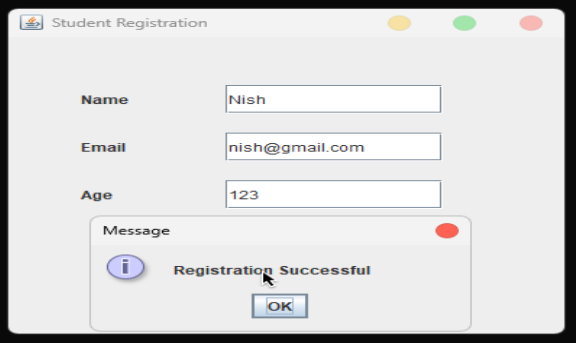
15. Write a Swing application that changes the window theme using radio buttons.

```java
import javax.swing.*;
import java.awt.*;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;

class as_11_q_15 {
    public static void main(String[] args) {
        JFrame f = new JFrame( title: "Theme Change");
        JRadioButton r1 = new JRadioButton( text: "Light");
        JRadioButton r2 = new JRadioButton( text: "Dark");
        ButtonGroup bg = new ButtonGroup();
        bg.add(r1);
        bg.add(r2);
        r1.setBounds( x: 100, y: 80, width: 100, height: 30);
        r2.setBounds( x: 200, y: 80, width: 100, height: 30);
        r1.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {
                f.getContentPane().setBackground(Color.WHITE);
            }
        });
        r2.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {
                f.getContentPane().setBackground(Color.GRAY);
            }
        });
        f.add(r1);
        f.add(r2);
        f.setSize( width: 400, height: 300);
        f.setLayout(null);
        f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        f.setVisible(true);
    }
}
```



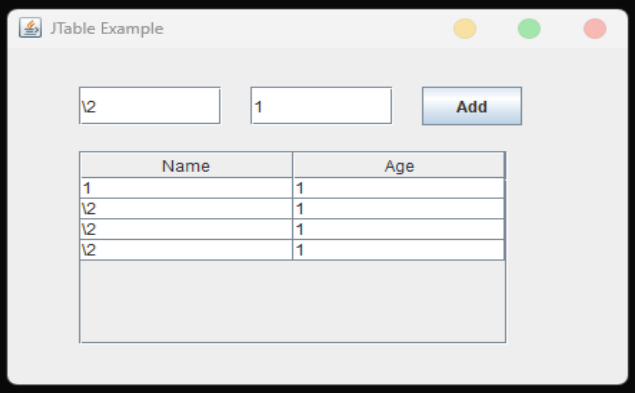
16. Create a Swing program that validates a student registration form (name, email, age).

```java
import javax.swing.*;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;

class as_11_q_16 {
    public static void main(String[] args) {
        JFrame f = new JFrame( title: "Student Registration");
        JLabel l1 = new JLabel( text: "Name");
        JLabel l2 = new JLabel( text: "Email");
        JLabel l3 = new JLabel( text: "Age");
        JTextField t1 = new JTextField();
        JTextField t2 = new JTextField();
        JTextField t3 = new JTextField();
        JButton b = new JButton( text: "Submit");
        l1.setBounds( x: 50, y: 50, width: 100, height: 30);
        t1.setBounds( x: 150, y: 50, width: 150, height: 30);
        l2.setBounds( x: 50, y: 100, width: 100, height: 30);
        t2.setBounds( x: 150, y: 100, width: 150, height: 30);
        l3.setBounds( x: 50, y: 150, width: 100, height: 30);
        t3.setBounds( x: 150, y: 150, width: 150, height: 30);
        b.setBounds( x: 120, y: 210, width: 100, height: 30);
        b.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {
                String name = t1.getText();
                String email = t2.getText();
                String age = t3.getText();
                if (name.equals("") || !email.contains("@")) {
                    JOptionPane.showMessageDialog(f, message: "Invalid Input");
                } else {
                    try {
                        Integer.parseInt(age);
                        JOptionPane.showMessageDialog(f, message: "Registration Successful");
                    } catch (Exception ex) {
                        JOptionPane.showMessageDialog(f, message: "Age must be number");
                    }}}});
        f.add(l1);f.add(t1);f.add(l2);f.add(t2);f.add(l3);f.add(t3);f.add(b);
        f.setSize( width: 400, height: 350);f.setLayout(null);
        f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);f.setVisible(true);
    }}
```



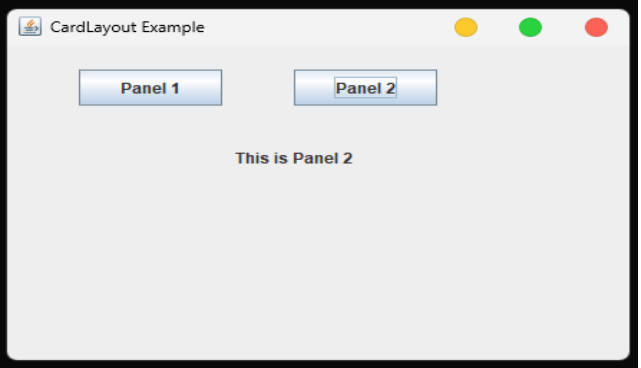
17. Design a GUI that stores user input in a table (JTable) and displays it dynamically.

```java
import javax.swing.*;
import javax.swing.table.DefaultTableModel;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;

class as_11_q_17 {
    public static void main(String[] args) {
        JFrame f = new JFrame( title: "JTable Example");
        JTextField t1 = new JTextField();
        JTextField t2 = new JTextField();
        JButton b = new JButton( text: "Add");
        DefaultTableModel model = new DefaultTableModel();
        model.addColumn( columnName: "Name");
        model.addColumn( columnName: "Age");
        JTable table = new JTable(model);
        JScrollPane sp = new JScrollPane(table);
        t1.setBounds( x: 50, y: 30, width: 100, height: 30);
        t2.setBounds( x: 170, y: 30, width: 100, height: 30);
        b.setBounds( x: 290, y: 30, width: 70, height: 30);
        sp.setBounds( x: 50, y: 80, width: 300, height: 150);
        b.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {
                model.addRow(new Object[]{
                        t1.getText(),
                        t2.getText()
                });}});
        f.add(t1);
        f.add(t2);
        f.add(b);
        f.add(sp);
        f.setSize( width: 450, height: 300);
        f.setLayout(null);
        f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        f.setVisible(true);
    }
}
```



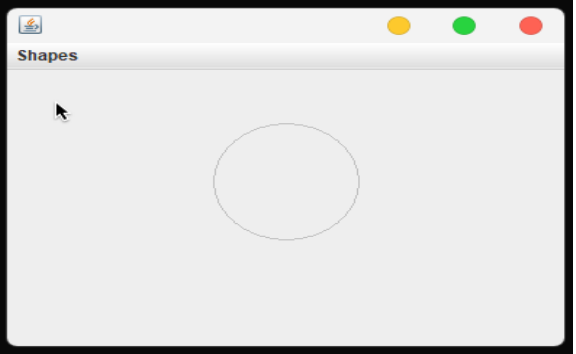
18. Write a program where clicking different buttons displays different panels using CardLayout.

```java
import javax.swing.*;
import java.awt.*;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;

class as_11_q_18 {
    public static void main(String[] args) {
        JFrame f = new JFrame( title: "CardLayout Example");
        CardLayout cl = new CardLayout();
        JPanel cards = new JPanel(cl);
        JPanel p1 = new JPanel();
        JPanel p2 = new JPanel();
        p1.add(new JLabel( text: "This is Panel 1"));
        p2.add(new JLabel( text: "This is Panel 2"));
        cards.add(p1, constraints: "1");
        cards.add(p2, constraints: "2");
        JButton b1 = new JButton( text: "Panel 1");
        JButton b2 = new JButton( text: "Panel 2");
        b1.setBounds( x: 50, y: 20, width: 100, height: 30);
        b2.setBounds( x: 200, y: 20, width: 100, height: 30);
        cards.setBounds( x: 50, y: 80, width: 300, height: 150);
        b1.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {
                cl.show(cards, name: "1");
            }});
        b2.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {
                cl.show(cards, name: "2");
            }});
        f.add(b1);f.add(b2);f.add(cards);
        f.setSize( width: 450, height: 300);f.setLayout(null);f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        f.setVisible(true);
    }
}
```



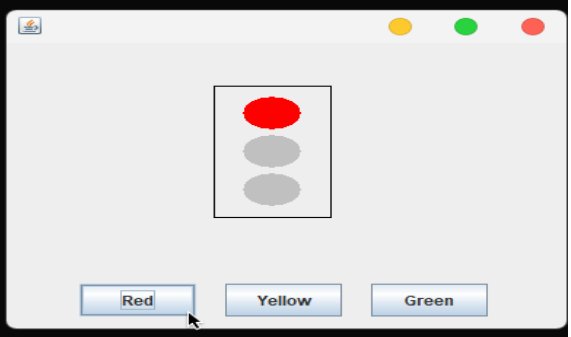
19. Design a GUI that draws simple shapes (circle, rectangle, line)when selected from a menu.

```java
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;
class as_11_q_19 extends JFrame {
    static String shape = "";  no usages
    as_11_q_19() {  no usages
        JMenuBar mb = new JMenuBar();
        JMenu m = new JMenu( s: "Shapes");
        JMenuItem c = new JMenuItem( text: "Circle");
        JMenuItem r = new JMenuItem( text: "Rectangle");
        JMenuItem l = new JMenuItem( text: "Line");
        c.addActionListener( ActionEvent e -> {
            shape = "circle";
            repaint();});
        r.addActionListener( ActionEvent e -> {
            shape = "rectangle";
            repaint();});
        l.addActionListener( ActionEvent e -> {
            shape = "line";
            repaint();});
        m.add(c);m.add(r);m.add(l);mb.add(m);
        setJMenuBar(mb);
        setSize( width: 400, height: 300);
        setLayout(null);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setVisible(true);}
    public void paint(Graphics g) {
        super.paint(g);
        if(shape.equals("circle"))
            g.drawOval( x: 150, y: 100, width: 100, height: 100);
        if(shape.equals("rectangle"))
            g.drawRect( x: 150, y: 100, width: 120, height: 80);
        if(shape.equals("line"))
            g.drawLine( x1: 100, y1: 150, x2: 300, y2: 150);
    }
    public static void main(String[] args) {
        new as_11_q_19();
    }
}
```



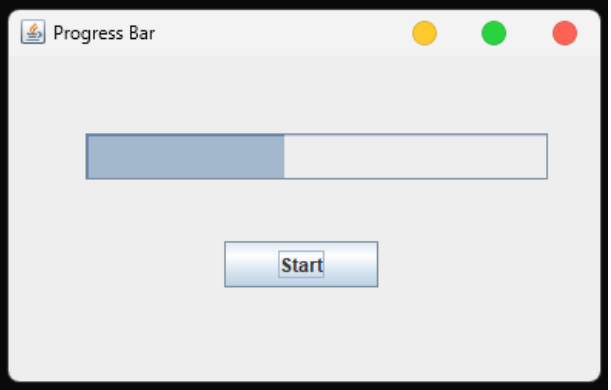
20. Write a Swing application that simulates a traffic signal using colored circles.

```java
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;
class as_11_q_20 extends JFrame {
    static String light = "";  no usages
    as_11_q_20() {  no usages
        JButton b1 = new JButton( text: "Red");
        JButton b2 = new JButton( text: "Yellow");
        JButton b3 = new JButton( text: "Green");
        b1.setBounds( x: 50, y: 220, width: 80, height: 30);
        b2.setBounds( x: 150, y: 220, width: 80, height: 30);
        b3.setBounds( x: 250, y: 220, width: 80, height: 30);
        b1.addActionListener( ActionEvent e -> {
            light = "red";
            repaint();});
        b2.addActionListener( ActionEvent e -> {
            light = "yellow";
            repaint();});
        b3.addActionListener( ActionEvent e -> {
            light = "green";
            repaint();});
        add(b1);add(b2);add(b3);
        setSize( width: 400, height: 300);
        setLayout(null);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setVisible(true);}
    public void paint(Graphics g) {
        super.paint(g);
        g.setColor(Color.BLACK);
        g.drawRect( x: 150, y: 70, width: 80, height: 120);
        g.setColor(light.equals("red") ? Color.RED : Color.LIGHT_GRAY);
        g.fillOval( x: 170, y: 80, width: 40, height: 30);
        g.setColor(light.equals("yellow") ? Color.YELLOW : Color.LIGHT_GRAY);
        g.fillOval( x: 170, y: 115, width: 40, height: 30);
        g.setColor(light.equals("green") ? Color.GREEN : Color.LIGHT_GRAY);
        g.fillOval( x: 170, y: 150, width: 40, height: 30);}
    public static void main(String[] args) {
        new as_11_q_20();
    }}
```



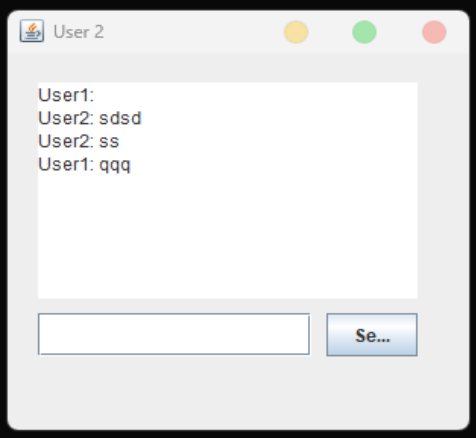
21. Develop a multi-threaded Swing program where a progress bar updates while a background task runs.

```java
import javax.swing.*;
class as_11_q_21 {
    public static void main(String[] args) {
        JFrame f = new JFrame( title: "Progress Bar");
        JProgressBar pb = new JProgressBar(0, 100);
        JButton b = new JButton( text: "Start");
        pb.setBounds( x: 50, y: 50, width: 300, height: 30);
        b.setBounds( x: 140, y: 120, width: 100, height: 30);
        b.addActionListener( ActionEvent e -> {
            new Thread(() -> {
                for (int i = 0; i <= 100; i++) {
                    pb.setValue(i);
                    try {
                        Thread.sleep( millis: 100);
                    } catch (Exception ex) {
                    }}
            }).start();});
        f.add(pb);f.add(b);
        f.setSize( width: 400, height: 250);
        f.setLayout(null);
        f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        f.setVisible(true);
    }
}
```



22. Design a Swing application that creates a real-time chat interface between two windows.

```java
import javax.swing.*;
class as_11_q_22 {
    public static void main(String[] args) {
        JFrame f1 = new JFrame( title: "User 1");
        JFrame f2 = new JFrame( title: "User 2");
        JTextArea a1 = new JTextArea();
        JTextArea a2 = new JTextArea();
        JTextField t1 = new JTextField();
        JTextField t2 = new JTextField();
        JButton b1 = new JButton( text: "Send");
        JButton b2 = new JButton( text: "Send");
        a1.setBounds( x: 20, y: 20, width: 250, height: 150);
        t1.setBounds( x: 20, y: 180, width: 180, height: 30);
        b1.setBounds( x: 210, y: 180, width: 60, height: 30);
        a2.setBounds( x: 20, y: 20, width: 250, height: 150);
        t2.setBounds( x: 20, y: 180, width: 180, height: 30);
        b2.setBounds( x: 210, y: 180, width: 60, height: 30);
        b1.addActionListener( ActionEvent e -> {
            String msg = t1.getText();
            a1.append("User1: " + msg + "\n");
            a2.append("User1: " + msg + "\n");
            t1.setText("");});
        b2.addActionListener( ActionEvent e -> {
            String msg = t2.getText();
            a2.append("User2: " + msg + "\n");
            a1.append("User2: " + msg + "\n");
            t2.setText("");});
        f1.add(a1);f1.add(t1);f1.add(b1);f2.add(a2);f2.add(t2);f2.add(b2
        f1.setSize( width: 320, height: 300);
        f2.setSize( width: 320, height: 300);
        f1.setLayout(null);f2.setLayout(null);
        f1.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        f2.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        f1.setVisible(true);f2.setVisible(true);
    }}
```

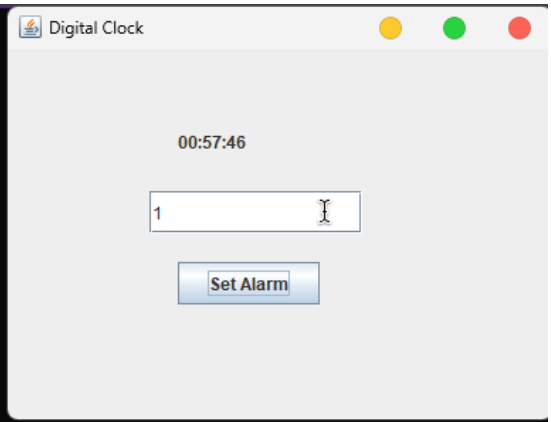


23. Write a program that visualizes sorting algorithms using animated bars.

```java
import javax.swing.*;
import java.awt.*;
class as_11_q_23 extends JFrame {
    int arr[] = {100, 50, 150, 80, 120};  no usages
    as_11_q_23() {  no usages
        JButton b = new JButton( text: "Sort");
        b.setBounds( x: 150,  y: 250,  width: 100,  height: 30);
        b.addActionListener( ActionEvent e -> {
            new Thread(() -> sort()).start();});add(b);
        setSize( width: 400, height: 350);
        setLayout(null);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setVisible(true);}
    void sort() {  no usages
        for(int i=0; i<arr.length-1; i++) {
            for(int j=0; j<arr.length-1-i; j++) {
                if(arr[j] > arr[j+1]) {
                    int temp = arr[j];
                    arr[j] = arr[j+1];
                    arr[j+1] = temp;
                    repaint();
                    try {
                        Thread.sleep( millis: 500);
                    } catch(Exception ex) {}}}}}
    public void paint(Graphics g) {
        super.paint(g);
        for(int i=0; i<arr.length; i++) {
            g.fillRect( x: 50 + i*60,  y: 220-arr[i],  width: 40, arr[i]);}}
    public static void main(String[] args) {
        new as_11_q_23();}}
```

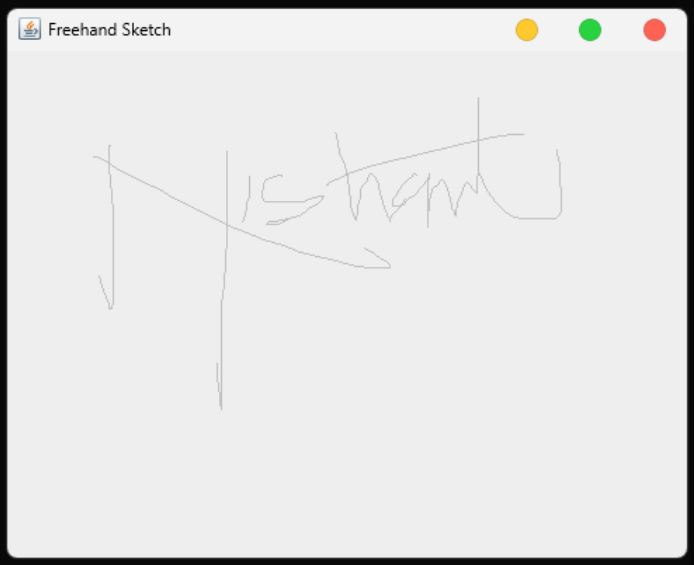
24. Create a GUI that simulates a simple digital clock with alarm functionality.

```java
import javax.swing.*;
import java.time.*;
class as_11_q_24 {
    public static void main(String[] args) {
        JFrame f = new JFrame( title: "Digital Clock");
        JLabel l = new JLabel();
        JTextField t = new JTextField();
        JButton b = new JButton( text: "Set Alarm");
        l.setBounds( x: 120,  y: 50,  width: 200,  height: 30);
        t.setBounds( x: 100,  y: 100,  width: 150,  height: 30);
        b.setBounds( x: 120,  y: 150,  width: 100,  height: 30);
        final String[] alarm = {""};
        b.addActionListener( ActionEvent e -> {
            alarm[0] = t.getText();});
        Timer timer = new Timer( delay: 1000,  ActionEvent e -> {
            String now = LocalTime.now().withNano( nanoOfSecond: 0).toString();
            l.setText(now);
            if(now.equals(alarm[0])) {
                JOptionPane.showMessageDialog(f,  message: "Alarm Time Reached!");
            }});
        timer.start();
        f.add(l);f.add(t);f.add(b);
        f.setSize( width: 400, height: 300);
        f.setLayout(null);f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        f.setVisible(true);}}
```



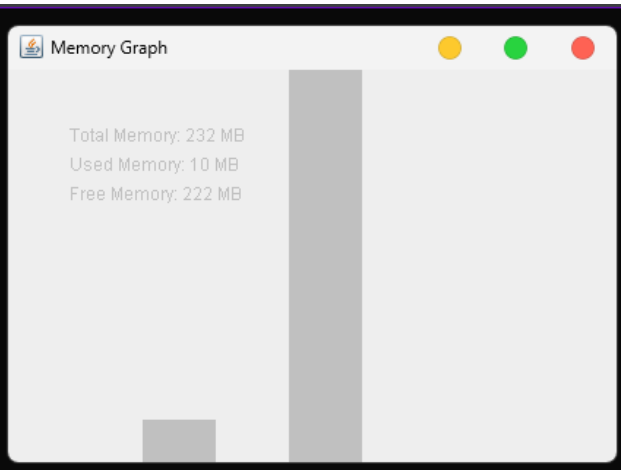
25. Develop a Swing application that draws freehand sketches using mouse drag events.

```java
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;
class as_11_q_25 extends JFrame implements MouseMotionListener, MouseListener {
    int x1, y1, x2, y2;  no usages
    as_11_q_25() {  no usages
        addMouseMotionListener( l: this);
        addMouseListener( l: this);
        setSize( width: 500, height: 400);
        setTitle("Freehand Sketch");
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setVisible(true);}
    public void mousePressed(MouseEvent e) {
        x1 = e.getX();
        y1 = e.getY();}
    public void mouseDragged(MouseEvent e) {
        x2 = e.getX();
        y2 = e.getY();
        Graphics g = getGraphics();
        g.drawLine(x1, y1, x2, y2);
        x1 = x2;
        y1 = y2;}
    public void mouseMoved(MouseEvent e) {}
    public void mouseReleased(MouseEvent e) {}
    public void mouseClicked(MouseEvent e) {}
    public void mouseEntered(MouseEvent e) {}
    public void mouseExited(MouseEvent e) {}
    public static void main(String[] args) {
        new as_11_q_25();}}
```



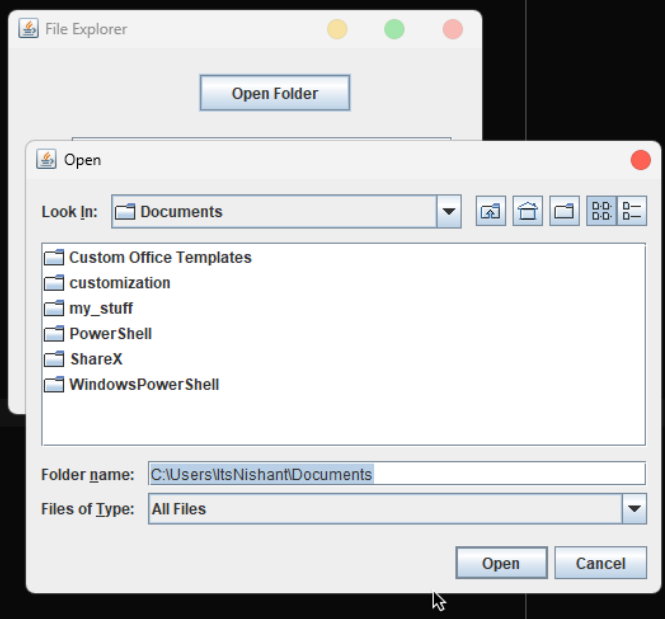
26. Write a program that displays system resource information (CPU, memory) in graphical form.

```java
import javax.swing.*;
import java.awt.*;
class as_11_q_26 extends JFrame {
    as_11_q_26() {  no usages
        setSize( width: 500, height: 400);
        setTitle("Memory Graph");
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setVisible(true);}
    public void paint(Graphics g) {
        super.paint(g);
        Runtime r = Runtime.getRuntime();
        long total = r.totalMemory() / 1000000;
        long free = r.freeMemory() / 1000000;
        long used = total - free;
        int usedHeight = (int)used * 3;
        int freeHeight = (int)free * 3;
        g.drawString( str: "Total Memory: " + total + " MB", x: 50, y: 80);
        g.drawString( str: "Used Memory: " + used + " MB", x: 50, y: 100);
        g.drawString( str: "Free Memory: " + free + " MB", x: 50, y: 120);
        g.fillRect( x: 100, y: 300-usedHeight, width: 50, usedHeight);
        g.fillRect( x: 200, y: 300-freeHeight, width: 50, freeHeight);
        g.drawString( str: "Used", x: 105, y: 320);
        g.drawString( str: "Free", x: 205, y: 320);}
    public static void main(String[] args) {
        new as_11_q_26();}}
```



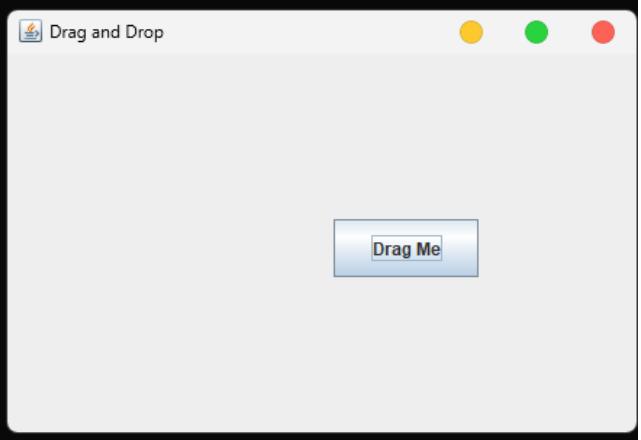
27. Create a GUI-based file explorer that lists files and folders from a directory.

```java
import javax.swing.*;
import java.io.File;
class as_11_q_27 {
    public static void main(String[] args) {
        JFrame f = new JFrame( title: "File Explorer");
        JButton b = new JButton( text: "Open Folder");
        DefaultListModel<String> model = new DefaultListModel<>();
        JList<String> list = new JList<>(model);
        JScrollPane sp = new JScrollPane(list);
        b.setBounds( x: 150,  y: 20,  width: 120,  height: 30);
        sp.setBounds( x: 50,  y: 70,  width: 300,  height: 200);
        b.addActionListener( ActionEvent e -> {
            JFileChooser fc = new JFileChooser();
            fc.setFileSelectionMode(JFileChooser.DIRECTORIES_ONLY);
            int result = fc.showOpenDialog(f);
            if (result == JFileChooser.APPROVE_OPTION) {
                File folder = fc.getSelectedFile();
                model.clear();
                File[] files = folder.listFiles();
                for (File file : files) {
                    model.addElement(file.getName());
                }}});
        f.add(b);f.add(sp);
        f.setSize( width: 450,  height: 350);f.setLayout(null);
        f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        f.setVisible(true);
    }
}
```



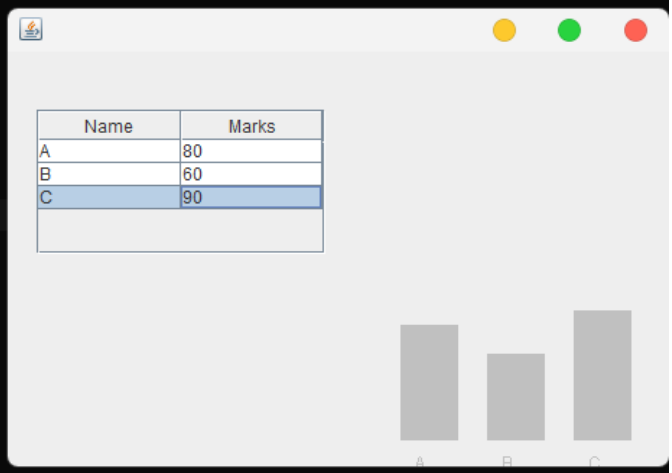
28. Develop a Swing application that allows users to drag and drop components within the window.

```java
import javax.swing.*;
import java.awt.event.*;
class as_11_q_28 {
    public static void main(String[] args) {
        JFrame f = new JFrame( title: "Drag and Drop");
        JButton b = new JButton( text: "Drag Me");
        b.setBounds( x: 150,  y: 100,  width: 100,  height: 40);
        final int[] x = new int[1];
        final int[] y = new int[1];
        b.addMouseListener(new MouseAdapter() {
            public void mousePressed(MouseEvent e) {
                x[0] = e.getX();
                y[0] = e.getY();
            }});
        b.addMouseMotionListener(new MouseMotionAdapter() {
            public void mouseDragged(MouseEvent e) {
                int newX = b.getX() + e.getX() - x[0];
                int newY = b.getY() + e.getY() - y[0];
                b.setLocation(newX, newY);
            }});
        f.add(b);
        f.setSize( width: 450,  height: 300);f.setLayout(null);
        f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        f.setVisible(true);
    }}
```



29. Write a Swing program that creates a multi-tab text editor using JTabbedPane.

```java
import javax.swing.*;
class as_11_q_29 {
    public static void main(String[] args) {
        JFrame f = new JFrame("Multi Tab Editor");
        JTabbedPane tp = new JTabbedPane();
        JTextArea t1 = new JTextArea();
        JTextArea t2 = new JTextArea();
        JTextArea t3 = new JTextArea();
        tp.add("Tab 1", new JScrollPane(t1));
        tp.add("Tab 2", new JScrollPane(t2));
        tp.add("Tab 3", new JScrollPane(t3));
        f.add(tp);
        f.setSize(500, 400);
        f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        f.setVisible(true);
    }}
```

30. Design a Swing application that generates a small dashboard displaying student statistics using charts and tables.

```java
import javax.swing.*;
import java.awt.*;
class as_11_q_30 extends JFrame {
    String names[] = {"A", "B", "C"};  no usages
    int marks[] = {80, 60, 90};  no usages
    as_11_q_30() {  no usages
        String data[][] = {
                {"A", "80"}, {"B", "60"}, {"C", "90"}};
        String col[] = {"Name", "Marks"};
        JTable table = new JTable(data, col);
        JScrollPane sp = new JScrollPane(table);
        sp.setBounds(20, 40, 200, 100);
        add(sp);
        setSize(500, 400);
        setLayout(null);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setVisible(true);}
    public void paint(Graphics g) {
        super.paint(g);
        for(int i=0; i<marks.length; i++) {
            g.fillRect(280 + i*60, 300-marks[i], 40, marks[i]);
            g.drawString(names[i], 290 + i*60, 320);}}
    public static void main(String[] args) {
        new as_11_q_30();}}
```